

Axial Potential Profile in the Wisconsin HTS Axisymmetric Mirror

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Github Repository

<https://github.com/tbpeterson000/Physics525-PlasmaExplorationProject-AxialPotential->

Introduction

This project aims to define the one-dimensional electrostatic axial potential profile in the Wisconsin HTS Axisymmetric Mirror (WHAM). WHAM is a linear magnetic mirror fusion device designed to have a maximum field strength of 17 tesla. However, the device currently possesses a reduced mirror ratio with a maximum field strength of approximately 8 tesla. The reduced magnetic regime is used throughout this project.

I am actively involved in research at WHAM investigating the application of a direct energy conversion device. This device will be placed in the expander region of WHAM converting the kinetic energy of escaping ions into usable DC power. This is primarily done by decelerating ions with a biased collecting plate before they are deposited on said plate. Through this procedure ions do work on the electric field, transferring their energy to an external circuit.

Fundamental to this research is having an accurate understanding of particle flux in the expander and the consequential potential. Thus this computational project aims to model particle behavior as they approach the bore and as they escape, moving in the expander.

Physics

This project is based on classical gasdynamic treatment of the mirror under the assumption of a collisional central cell. This simply serves to define the source of particles, as no fluid modeling was conducted. Ions are released inside the central cell from approximately an ion-mean-free-path away from the mirror bore. The initial velocity components of the ions are specific by sampling from a thermal/Maxwellian distribution with a constant temperature of 100 eV. The goal here is to simulate collisionless dynamics of ions as they move toward the bore. While many ions are reflected back into the central cell, some will go through the mirror into the expander. Thus this study involves both trapped and non-trapped particles. Tracking the particle

trajectories allows for a calculation of the z dependence of ion density between the release point and the bore. The trajectory of each particle is calculated using the Lorentz force equation, first with only the magnetic term.

$$m \frac{dv}{dt} = qv \times B$$

The calculation of density is reliant on the assumption of negligible radial variation of density in the area of interest. A flux tube was defined along the magnetic field lines, having a diameter of 2 centimeters at the release point. There is little expansion of the field lines in this domain, allowing this assumption. Under the assumption of quasineutrality while still inside the bore, the axial ion density can be equated to the axial electron density.

$$n_i(z) = n_e(z)$$

With the previous definitions, the axial electron density can be converted to the axial electrostatic potential through the Boltzmann Relation.

$$\phi(z) = T_{eV} \ln\left(\frac{n_e}{n_0}\right)$$

This generates an electrostatic potential profile for the area between the release point and the bore. This solution however, does not take into account the electrodynamic interaction between species. To obtain a more accurate solution, the model is iterated by feeding the calculated electrostatic potential back into the simulation through the addition of the electric component of the Lorentz force.

$$m \frac{dv}{dt} = q(E + v \times B)$$

Conducting this iteration converges the produced electrostatic potential to a self-consistent, steady-state solution.

Methodology

This project was conducted using Comsol Multiphysics. The first step in the project was defining a stationary study to calculate the static magnetic field. This is done by specifying WHAM's bore and central field coils using a geometry feature. Applying a magnetic field physics node allows for the addition of coil features, in which the material properties and current within each coil can be set. In order to obtain a free space solution, an infinite element domain was applied to an outer shell of the magnetic field boundary. This circumvents Comsol's automatic reinforcement of tangential fields at boundaries. Mesh refinement was conducted on the bore magnets to ensure

more accurate calculation of the magnetic field where the field lines are compressed. The results of this refinement can be seen in the uniform distribution of field lines at the bore in figure 1b.

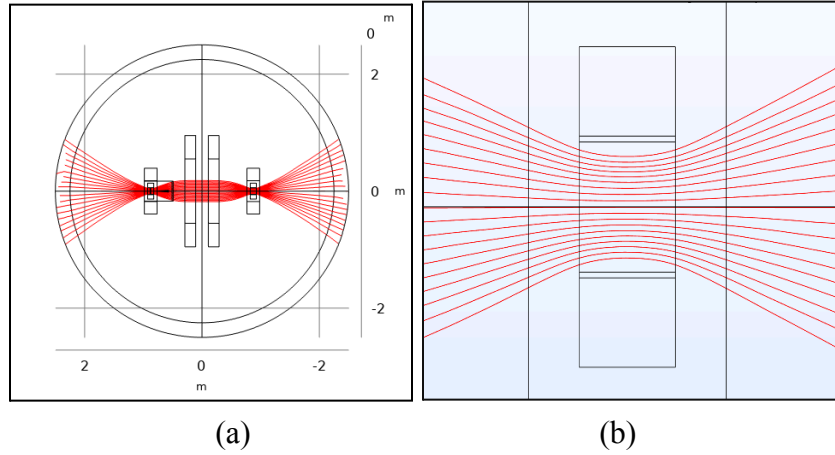


Figure 1: Calculated Magnetic field lines given WHAM coil parameters. (a) displays the results of the infinite element domain while (b) demonstrates uniform field lines at the bore.

The next step involves defining a charged particle tracing feature and adding it to a new time-dependent study step. To specify the flux tube in which particle tracing is done, a streamline of the magnetic field, initiated 1 centimeter off axis at the z coordinate of the source, is exported to SolidWorks. There the field line can be revolved into the desired flux tube geometry. This volume can then be imported back into Comsol, defining the flux tube. The procedure is repeated for a flux tube shell of thickness 1 centimeter to account for particles whose larmor orbit periodically escapes the principal flux tube. A cylindrical volume is defined around the flux tube with ample separation to serve as particle termination walls for the particle trajectories. The mesh elements are refined to maximize accuracy in trajectory resolution, as seen in Figure 2.

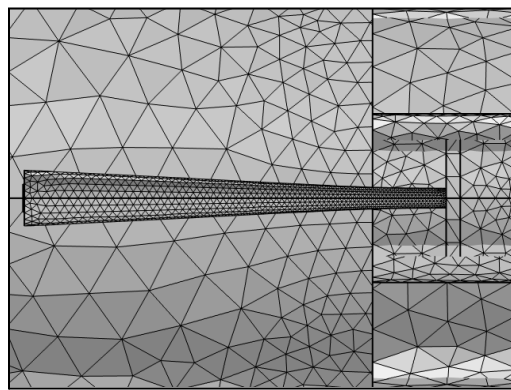


Figure 2: Mesh refinement of defined flux tube. (Maximum element size 0.13m, minimum element size 0.001m, maximum element growth rate 1.35)

To initiate particles, an inlet is defined at the source face of the flux tube. The initial position of particles on this face is specified using a uniform density. The initial velocity is specified using

the thermal option and a user defined temperature of $100 \text{ eV} = 1.16 \times 10^6 \text{ K}$. Particle species are specified as protons, corresponding to an H_2 plasma. A density accumulator is added to the flux tube domain in the charged particle trajecting feature. This provides the accumulated density across the flux tube over the time interval. In the study node, the stationary step is defined for the magnetic field calculation and the time dependent step is defined for the particle tracing. The particle tracing uses the field calculated in the stationary step for the magnetic force parameters.

Once the study is completed, volume integrals are defined within the derived values section of the results node. These integrals use if statements to calculate the integrals the density and volume for 2 centimeter slices. The calculated values are then exported to Excel files for postprocessing.

A python script was written to pull the calculated density and volume integrals. The script first divides the density integral, which is a number of particles, by the corresponding volume integral. This produces the average density within each slice. Then the Boltzmann relation defined in the above physics section is applied to each density slice. On the first iteration this raw potential profile is then exported to another Excel file.

Back in Comsol, this file is called into a Least-Squares Fit feature to fit an even 6th order polynomial to the potential. The resultant curve is then fed back into the study by activating an electric force feature within the charged particle tracing node. Repeating the study now accounts for the electrostatic potential and iterations can be made.

When the python script is executed with previous runs in the Excel file, a 25% relaxation coefficient is applied to the new potential and added to 75% of the old potential. This serves to slow down the convergence, ensuring any large variation does not blow up the potential. A plot showing the new potential and the calculated relaxed potential is generated.

Results

The study was run with 15,000 particles and a time domain of $[0, 2^{-5}\text{s}]$ with output intervals every 5^{-7}s . The accumulated particle density sampled along the axis, and the binned axial density from postprocessing can be seen in Figure 3. Note the magnitude of noise in figure 3a, motivating the binning process. The large density spike seen in figure 3a at $\sim 0.25\text{m}$ is averaged out across the slice volume, generating the clean curve shown in figure 3b.

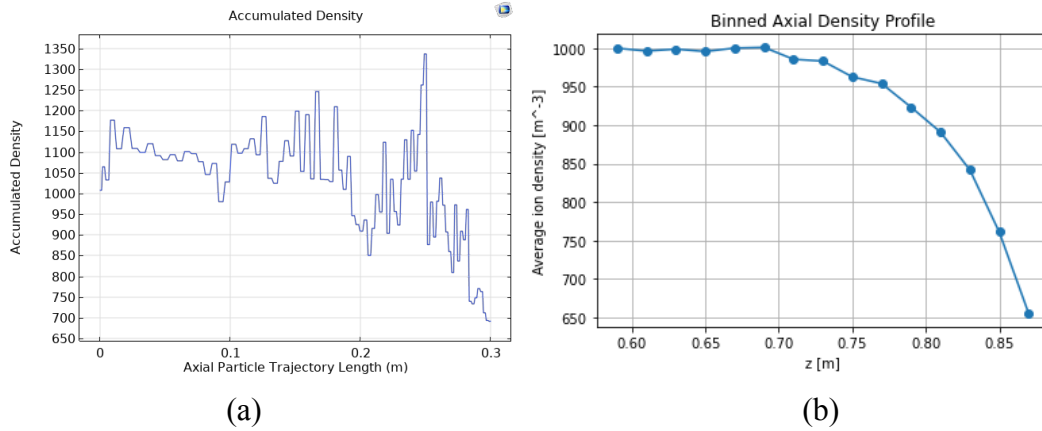


Figure 3: Plot (a) displays the accumulated particle density sampled along the axis. Plot (b) displays the corresponding binned density results.

Applying the Boltzmann relation through the python script generates the curve shown in figure 4a. For iterations the raw vs relaxed electrostatic potential can be seen in figure 4b. Figure 4c illustrates the change in potential at each binned value across iterations, demonstrating convergence of the model.

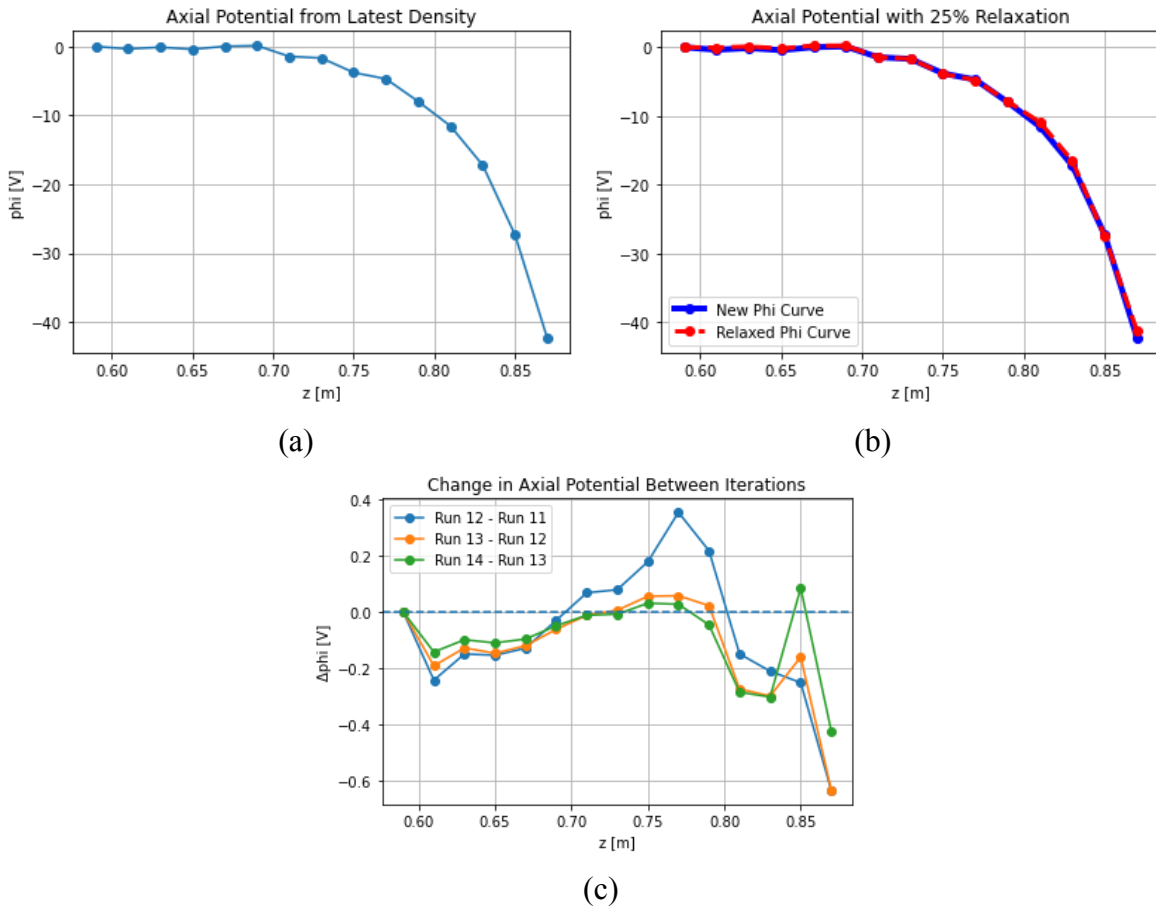


Figure 4: Plot (a) displays the computed axial potential. Plot (b) shows the electrostatic potential curves for iteration. Plot (c) shows the change in potential between iterations.

Completing iterations until reasonable convergence has been achieved produces figure 5. This represents the change in axial electrostatic potential between the release point and the bore. An even, 6th-order polynomial is fit to the points while reinforcing intersection with the first value. The fitted expression is seen below with coefficients rounded to two decimal places.

$$\phi(z) = -86219.07(z - 0.59)^6 + 1206.41(z - 0.59)^4 - 87.16(z - 0.59)^2$$

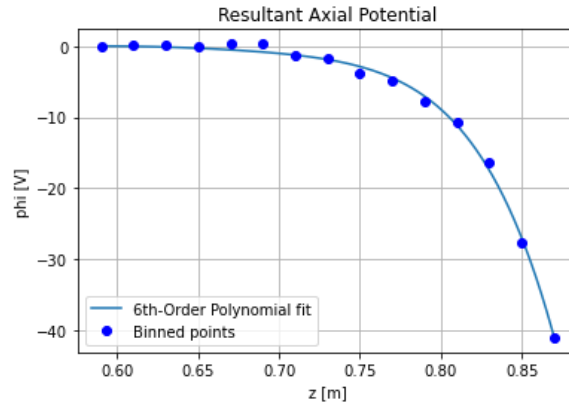


Figure 5: Final calculated electrostatic potential.

Interpretation

As electrons move to escape through the bore a separation of charge builds. This in turn generates an electric field to prevent such separation. The decreasing electrostatic potential continues to build across the expander. Electrons are reflected by the intensifying negative potential and pulled back into the central cell. In this way, ions are primarily confined by the mirror effect while electrons are confined by the developed potential. Ions that are able to escape the mirror will be accelerated by this electrostatic potential before colliding with an endplate. Thus, the potential contributes to defining the ions final energy before hitting the collector.

The work done here can be evaluated based on the findings of Wetherton et al. Their study calculated the ambipolar potential drop within the mirror expander using advanced particle-in-cell methods. They show that the ambipolar potential forms self-consistently through quasineutrality and flux balance. Figure 6 displays a key plot from their paper, showing the axial electron density (blue) and the parallel potential (red). They report the potential from the central plane, through the bore, and to the expander. Looking at the region of their potential

corresponding to the region of interest of this work shows that they calculated a potential drop between the central cell and the bore of approximately $\frac{e\phi_{||}}{2T_0}$.

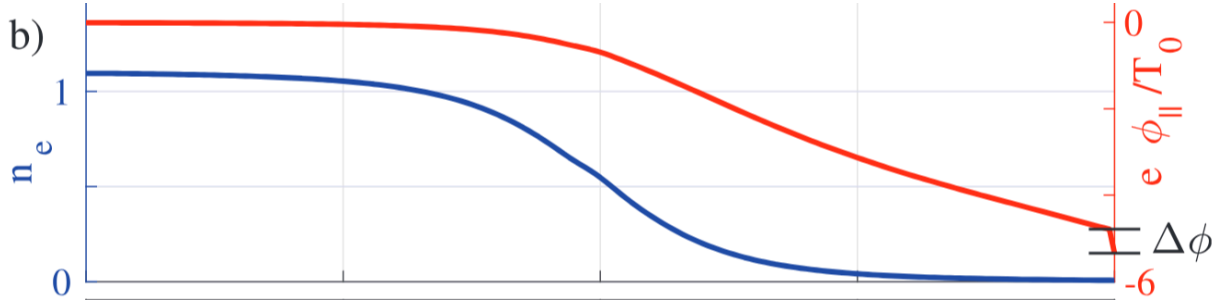


Figure 6: Electron density (blue) and ambipolar potential drop (red) from the middle plane of a magnetic mirror (left) to the endplate(right). Note the x axis is from 0 to L.

My results show an electrostatic potential drop from a source of collisionless plasma to the bore of the mirror of -40 V. As the designated value of T_i for this study was 100 eV, the expected potential drop according to the findings of Wetherton et al. is -50V. Therefore, considering the simplifying assumptions made in this computational project, the results obtained are in reasonable agreement with expected values.

Although the present approach does not solve for the fully self-consistent ambipolar potential through the expander, it does capture the electrostatic structure between the release point and the bore. The established agreement in magnitude and trend as outlined above suggests that this reduced model based on density and the Boltzmann relation adequately represents the dominant physics governing the axial potential.

Outlook

This computational project has represented an extremely intriguing part of my semester in Physics 525. I plan to continue my work with this study, specifically looking at defining the electrostatic potential across the expander with collisionless dynamics of both electrons and ions. As the Boltzmann relation no longer holds here, I aim to find the potential by solving the Poisson equation. Using the completed work, I will be able to estimate the density of ions at the bore. I will launch thermal ions from the preexisting release point alongside thermal electrons from the bore with half the calculated ion density there. This will represent a realistic energy distribution for both species. Following a similar procedure as this project, I will calculate the density of both species and feed them into the Poisson equation. This can again be iterated to create self-consistency. I look forward to the results of these studies.

References

Wetherton, B. A., A. Le, J. Egedal, et al. "A Drift Kinetic Model for the Expander Region of a Magnetic Mirror." *Physics of Plasmas* 28, no. 4 (2021): 042510.

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